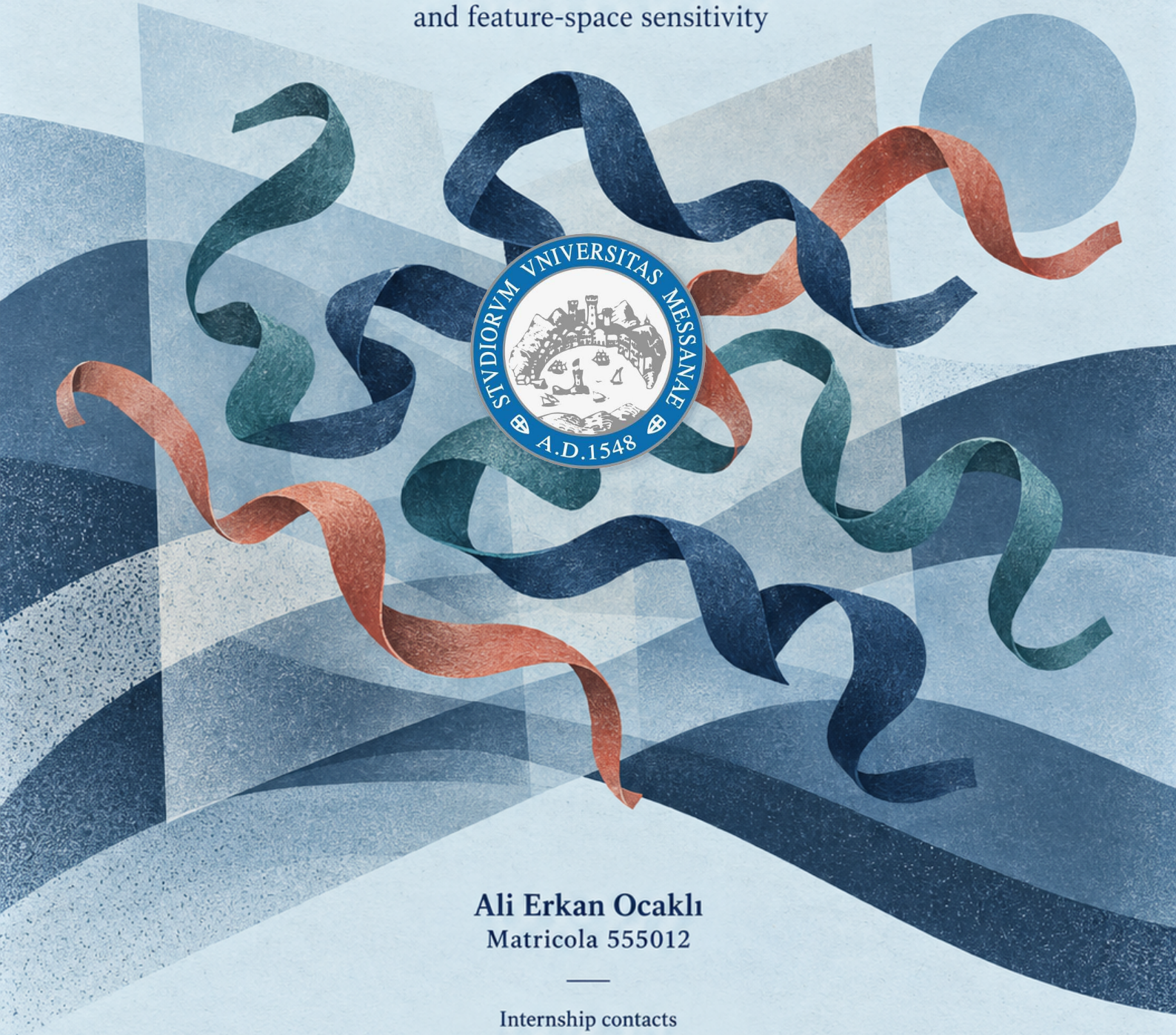
UNIVERSITY OF MESSINA
Data Analysis L-31

CURRICULAR INTERNSHIP REPORT

# RobustCluster-Aβ

## Reproducible analysis of amyloid-β40 conformational descriptors

Scientific code audit, sampling diagnostics and feature-space sensitivity




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# Abstract

The analysis of intrinsically disordered peptides requires a distinction between the configurations saved by a simulation, their equilibrium statistical weights, and the choices used to group them. This internship report develops RobustCluster-Aβ to examine these questions using amyloid-β40 descriptors supplied for reproduction of the workflow of Cadenelli and colleagues. The published approach combines well-tempered metadynamics, density-peak clustering, Wasserstein consensus, final assignment and statistical reweighting. The present project separates that reproduction target from corrections to the supplied script and an independent analysis of feature sensitivity.

The descriptor table contains 300,003 observations in three replica segments, each covering the source-labelled 1–3 microsecond interval. An audit identified problems involving parsing, residual clusters, free-energy interpolation, replica pooling and weighted summaries. The modular implementation preserves the original inputs and adds numerical tests, parameter files, input hashes and run manifests. A subsequently recovered free-energy surface passed grid-completeness and interpolation checks for every observation; its energy-unit and construction provenance remain unresolved.

Completed descriptor diagnostics show a strong radius-of-gyration/SASA correlation and a concentrated principal-component spectrum. Correlation-time reliability checks failed for all twelve replica–descriptor series, limiting interpretation of their provisional effective-sample-size estimates. An independent Gaussian-mixture experiment then compared four-feature partitions with partitions obtained after removing SASA. Fifty models were fitted using a fixed replica-balanced sample and five initializations at each of five component counts. All fits reported convergence. Agreement and seed stability varied with component count, showing that global feature correlation alone does not establish redundancy for clustering.

The report also documents two companion research-software projects, VMD MCP and GROMACS MCP, and connects their implementation choices to reproducible analysis. The computed results describe the biased descriptor sample; they do not establish equilibrium cluster populations, molecular structures or physical kinetic rates. The outcome is an executable, audited project with a completed original sensitivity study and an explicit path toward full reproduction.

**Keywords:** amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; density peaks; statistical reweighting; reproducibility.

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# Abbreviations and evidence categories

| Term | Meaning |
|---|---|
| ACF | Autocorrelation function |
| ARI | Adjusted Rand index |
| CV | Collective variable |
| GMM | Gaussian mixture model |
| DPC / DPA | Density Peak Clustering / Density Peaks Advanced |
| ESS | Effective sample size; estimator and assumptions stated explicitly |
| FES | Free-energy surface |
| IDP | Intrinsically disordered protein or peptide |
| MD | Molecular dynamics |
| PCA | Principal component analysis |
| SASA | Solvent-accessible surface area |
| SEM | Standard error of the mean |
| WT-MetaD | Well-tempered metadynamics |

Published methodology, the implemented reproduction, corrections to supplied code, robustness extensions, and future work are distinct evidence categories throughout this report. A successful software test demonstrates behavior on a specified fixture; it does not establish a biological result. A BLOCKED analysis lacks required inputs or provenance and is not presented as completed. The report distinguishes assigned learning topics, demonstrable project outputs, and analyses that remain unperformed.

# 1. Introduction and internship objectives

## 1.1. Biological and computational motivation

An intrinsically disordered peptide is described by an ensemble rather than a single reference structure. Configurations with similar global dimensions can differ in secondary structure, solvent exposure and contacts. Analysing amyloid-β monomers is therefore a problem of statistical representation as well as molecular simulation. Disorder does not imply an absence of structural organization [1].

The motivating study compares monomeric Aβ40 and Aβ42 ensembles using a common simulation and analysis protocol [2]. Only Aβ40 descriptors are available here. The present project investigates their analysis and reproducibility; it does not compute an isoform comparison or aggregation rates. Related energy-landscape studies motivate ensemble characterization without making monomeric clustering a direct measurement of fibril formation [3].

## 1.2. Internship context and objectives

This project forms part of the Data Analysis L-31 internship at the University of Messina, with Andrea Ciccolo and Valeria Conti Nibali as the contacts supplied by the student. The documented programme moved from protein structure and intrinsic disorder to force fields, molecular dynamics and clustering. Its final-stage brief emphasized an independent alternative method or analysis using knowledge from the degree.

Three objectives follow: audit the supplied inputs and script; implement the published workflow with explicit corrections; and investigate how representation affects an independently fitted partition. The central question is: how much of the observed grouping is supported by the descriptors, and how much depends on analysis choices? A Gaussian-mixture SASA-ablation experiment addresses that question without requiring equilibrium weights.

## 1.3. Contributions and evidence boundaries

Completed work includes source and data validation, a modular Python package, numerical tests, correlation/PCA and sampling diagnostics, complete-grid FES validation, and the independent feature-ablation pilot. A complete FES is now available, but its energy units and construction provenance still require clarification before statistical reweighting. Published methods, implementation corrections and original experiments are kept separate.

Each computation records input hashes, configuration, software versions and outputs. Figures are generated from those records; exploratory notebooks are not the source of truth. This follows the emphasis on portable workflows and machine-actionable provenance in research-software guidance [4–6]. The report connects these practices to the internship's learning objectives while distinguishing assigned study topics from completed numerical work.

# 2. Scientific and theoretical background

## 2.1. Amyloid-β40 and intrinsically disordered ensembles

For an IDP, an ensemble average combines contributions from many conformations with different statistical weights. A single representative structure is therefore an illustration of one configuration, not a replacement for the ensemble. Global descriptors provide a compact summary, but two conformations can share the same radius of gyration and solvent-accessible area while having different residue contacts. A cluster in descriptor space should consequently be interpreted as a family under a specified representation, not as a unique atomic fold.

Experimental constraints can improve the characterization of such ensembles, but observables such as NMR or SAXS measurements generally average over conformations. Borthakur and colleagues describe a maximum-entropy procedure that combines simulations with experimental data to constrain IDP ensembles [7]. That study motivates an external validation route for this project. It does not provide a license to infer missing structures or to equate differently defined observables without checking their compatibility.

## 2.2. Molecular dynamics and conformational ensembles

Molecular dynamics evolves a model system according to a chosen potential-energy function and numerical integration scheme. The resulting distribution depends on force-field parameters, solvent representation, thermodynamic conditions, and sampling. The primary paper reports GROMACS simulations using CHARMM36m and PLUMED [2, 8–10]. These are statements about the published study. The present project analyzes its supplied descriptors and has not rerun those molecular simulations.

In an equilibrium canonical ensemble, an observable $A(q)$ has expectation proportional to the integral of $A(q)\exp[-\beta_T U(q)]$, where $q$ denotes coordinates and $U$ is potential energy. For energies per molecule, $\beta_T = 1/(k_B T)$; for the molar energies used in the implementation, the consistent factor is $1/(RT)$. Mixing these energy conventions introduces a scale error even when the reweighting code is numerically stable.

## 2.3. Sampling limitations

A long sequence of saved configurations does not imply an equally large number of independent observations. Nearby frames may share a local environment or remain in slowly changing regions of descriptor space. Statistical inefficiency quantifies the variance inflation associated with serial correlation under suitable assumptions, but it cannot repair inadequate exploration. A stable mean can also conceal compensating changes in different structural subpopulations.

Assessing equilibration and sampling quality requires evidence beyond a raw frame count [11, 12]. Here, an additional limitation is that the trajectory was generated under a changing bias. Correlation statistics are calculated as diagnostics of that recorded process. They are not identified with unbiased relaxation times, transition rates, or kinetic states. The fixed published analysis interval is retained for reproduction; the extension assesses its statistical behavior rather than silently choosing a new interval.

## 2.4. Well-tempered metadynamics

WT-MetaD adds a history-dependent potential along selected CVs. Its tempering parameter is conventionally expressed as $\gamma = (T + \Delta T)/T > 1$. In the long-time regime, the bias approaches a scaled negative free energy, up to an additive constant [13]. This relationship motivates the final-bias approximation used in the primary paper.

The essential distinction is between accelerating exploration and preserving an unbiased distribution. Biased occupancies are not equilibrium populations. Moreover, the modified forces alter the connection between recorded trajectory time and physical dynamics. Thermodynamic reweighting can address

populations under its assumptions; it does not automatically restore kinetic information.

## 2.5. Gaussian hills and bias construction

For CV vector $s$, hills deposited at times $t_k$ generate a bias of the form

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{2.1}$$

In WT-MetaD, deposited heights decrease with the accumulated bias as $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B\Delta T)]$, using molecular energy units. For molar units, $k_B$ is replaced by $R$. The precise convention must agree with the simulation software and energy units. Equation 2.1 describes the methodological construction, not a reconstruction performed in this project. No project HILLS file is available.

PLUMED distinguishes the instantaneous bias output, the time-dependent offset, and their difference. Its `metad.rbias` component equals `metad.bias` minus `metad.rct`; the latter represents $c(t)$ [14]. A column named `metad.rbias` is therefore potentially useful for a separate time-dependent reweighting analysis. Its name alone does not establish that the offset was computed with the required settings or that its units are known.

## 2.6. Free-energy surfaces

For a CV distribution $p(s)$, free energy is $F(s) = -\beta_T^{-1} \log p(s) + C$, where $C$ sets an arbitrary reference. Free energy is a property of the chosen projection; integrating out other coordinates can obscure structural distinctions. A two-dimensional minimum does not uniquely specify a three-dimensional conformation.

PLUMED's `sum_hills` supports reconstruction from accumulated hill records and distinguishes free-energy output from negative-bias output [15]. That distinction matters because the WT scaling factor must not be applied twice. The recovered complete file labels its energy field `file.free`, but the energy unit, generating commands, and replica-combination procedure are still needed for a verified final-bias calculation.

## 2.7. Statistical reweighting

For a quasi-static bias, the correction from a biased distribution to the corresponding unbiased distribution has weights proportional to $\exp(\beta_T V)$. Combining this with the WT relationship yields

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2.2}$$

An additive constant in $F$ changes every log weight by the same amount within the chosen estimator and disappears after normalization. Equation 2.2 is the baseline implemented here, following the primary paper's final-bias approximation. Its scientific validity depends on the quasi-stationary interpretation of the analyzed interval.

Alternative MetaD reweighting approaches account explicitly for changing bias normalization [16, 17]. For verified PLUMED offset-corrected output, the optional comparison uses $\ell_i = \mathrm{rbias}_i/(RT)$ [18]. These two estimators are not interchangeable labels for the same computation. The current repository keeps their configurations and resulting outputs separate.

## 2.8. Structural descriptors

The baseline uses $n_\alpha$, $n_\beta$, $R_g$, and total SASA. In the primary study, the first two are scaled continuous secondary-structure CVs, intended to approximate residue counts. They should not be interpreted as exact integer DSSP assignments. DSSP itself classifies local structural patterns using geometrical and hydrogen-bond information [19]; reproducing such assignments requires coordinates, which are absent here.

Radius of gyration summarizes global compaction, whereas SASA describes solvent exposure. Their information can overlap without being identical. Additional columns describe apolar and polar SASA and intra- and peptide–solvent hydrogen bonds. These are characterized after clustering, not used to select baseline families. Their upstream definitions remain tied to the supplied provenance; in particular, apolar and polar areas are not assumed to sum exactly to total SASA.

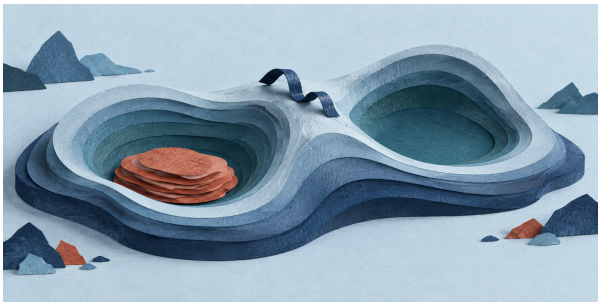
## 2.9. Density-based conformational clustering

Density-peak methods seek centers that combine high local density with separation from denser points. The primary study adapts that logic to recursively discover acceptable clusters across random subsets and density scales [2]. A silhouette coefficient summarizes the relation between within-cluster cohesion and separation from other clusters [20]. It is a geometric criterion, not proof of thermodynamic metastability.

Consensus then groups local clusters by similarities in their descriptor distributions. DPA offers a distinct, statistically motivated density-topography approach [21]. It is a possible benchmark, not the baseline implemented by substituting a library call. Likewise, weighted ShapeGMM addresses structural ensembles in a coordinate-based representation [22]; its applicability here depends on obtaining valid structural frames.



**(a)** Conformational diversity



**(b)** An evolving sampling landscape

**Figure 1.:** Conceptual editorial illustrations. Open paper ribbons evoke different conformations of a flexible peptide; layered deposits evoke the accumulation of metadynamics bias. Neither image depicts molecular coordinates, secondary-structure assignments, a measured FES or a physical trajectory. The qualitative ideas are formalized in the surrounding text and eq. (2.1). Generated-art provenance is documented in Appendix D.

# 3. Data and computational methodology

## 3.1. Data provenance

The input is a 10-column text table with 300 003 numerical rows. Its SHA256 and the hash of the original script are recorded in the inventory. The original materials are retained unchanged. The table is not an atomic trajectory: it contains scalar descriptors and an offset-corrected-bias-labelled column. It cannot support new contact maps or molecular renderings without an independently verified link to structural frames.

The local publisher PDF establishes the published method; the supplied code establishes implementation evidence. Neither is treated as a substitute for missing execution logs. Scientific V2 and its changelog were not located, so the repository does not claim to reconstruct their contents or authorship. The audit documents retain an explicit missing-evidence entry for that intermediate version.

## 3.2. Replica segmentation

Replica boundaries are detected where the first column decreases. Parsing requires exactly three segments, strictly increasing times within each segment, finite numerical entries, and a recognized header. Extra segments, duplicate times, malformed rows, or missing interval coverage fail validation rather than being silently skipped.

Each observation retains its original zero-based source row, physical one-based file line, replica ID, and zero-based frame index within the original segment. Those identifiers remain unchanged when the interval is selected or clustering groups are randomized. Stable frame identities are necessary for comparisons across seeds: comparing arrays solely by their current order could create a false disagreement between otherwise identical partitions.

## 3.3. Analysis interval

The baseline interval includes both endpoints from 1 000 000 to 3 000 000 in the source-labelled picosecond time unit. Each replica contributes 100 001 frames separated by 20 ps. The original script and published protocol support this unit interpretation, but the upstream simulation/analysis metadata are absent. The interval therefore corresponds to the supplied 1–3 μs window under a recorded provenance assumption.

The first microsecond is excluded in the published workflow. Because the supplied table already begins at the selected boundary, the present data cannot independently establish what happened before that interval or reproduce the paper's convergence assessment over the complete trajectory. The code verifies coverage of the requested interval; it does not invent earlier observations.

## 3.4. Descriptor definitions

The canonical clustering columns are $n_\alpha$, $n_\beta$, $R_g$ in nm, and SASA in $\mathrm{nm}^2$, with units inherited from the original script. Header spelling differences are mapped explicitly, including the original `HBONNDSPS` spelling. The `metad.rbias` column is preserved as a separate field and is not renamed to an interpolated free energy.

The CV coordinates for FES queries are $\alpha = n_\alpha/1.1$ and $\beta = n_\beta/2.5$. These transformations refer to the paper's descriptor scaling, not to min–max normalization. Applying the min–max feature transform before an FES lookup would query the wrong physical CV coordinates. The two transformations are therefore implemented in different modules and saved separately.

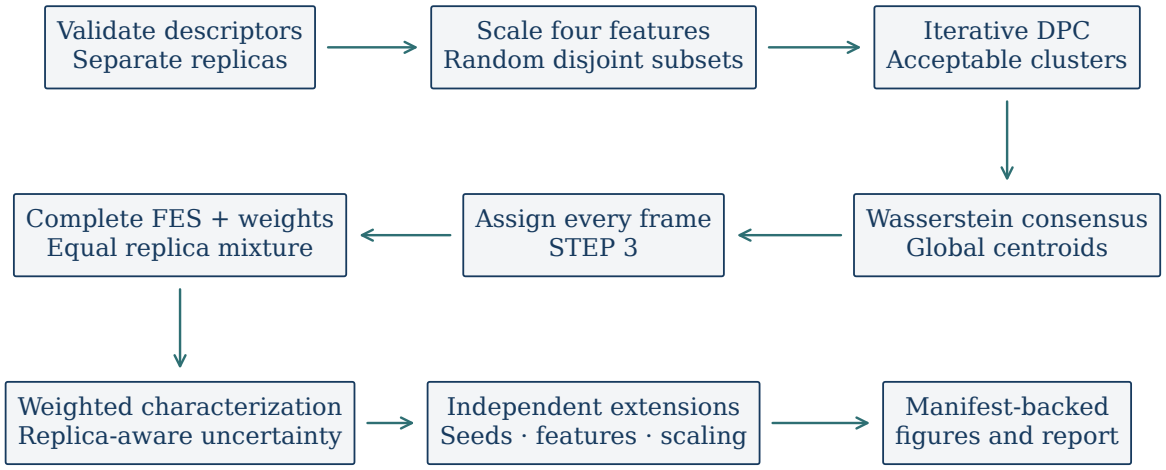



**Figure 2.:** Implemented analysis design. The primary branch separates geometric cluster discovery from statistical reweighting. Robustness experiments rerun designated stages independently. The FES grid has been validated, while empirical weighting remains blocked by energy-unit and construction provenance; this workflow diagram documents implementation, not successful completion of every box.

## 3.5. Feature normalization

For each baseline descriptor $x_f$, pooled analysis-window min–max scaling gives $z_f = (x_f - a_f)/(b_f - a_f)$, where $a_f$ and $b_f$ are the pooled minimum and maximum. Both values are saved. A constant feature is an invalid input for this transform, rather than a reason to divide by zero or silently remove a scientific variable.

Pooling defines a common metric for every subset and replica. Independent per-group scaling would make the same molecular observation occupy different numerical coordinates depending on its random group. The fixed transformation prevents that inconsistency. Min–max scaling is source-informed; sensitivity to a different metric is tested only in the extension.

## 3.6. STEP 1: iterative Density Peak Clustering

The workflow in fig. 2 begins with random disjoint subsets. Production group size is approximately 30 000. A tiny final subset of at most ten frames is redistributed over preceding groups, preserving every identity. Local RNG streams are derived from a recorded master seed. This removes the legacy fixed-seed-list limit, but changes exact random draws relative to the original global-RNG implementation [23].

The density estimate counts neighbors within $d_c$, including the point itself as in the supplied code. Density ties are ordered deterministically by local frame position. A point's separation $\delta_i$ is its distance to the closest point earlier in that ordering; the first point uses the maximum pairwise distance. Centers are ranked by $\rho_i\delta_i$. This tie convention is documented because equal cutoff densities are common, and a claim that every parent is strictly denser would be inaccurate.

For each cutoff, candidate center counts are scanned from 10 down to 2. The selected candidate maximizes the number of acceptable clusters; ties retain the earlier, higher-center-count candidate. Acceptance requires at least 300 members and mean silhouette at least 0.3. After accepted members are removed, $d_c$ is multiplied by 0.9. The process stops when fewer than 300 points remain or five consecutive iterations accept no cluster. Residual points remain unassigned at this stage.

Distances are computed in bounded blocks into a float32 matrix. Cutoff density is evaluated row by row without an additional floating kernel matrix. Silhouettes reuse that distance matrix, subject to the validity

condition that the number of labels lies between two and $N - 1$ [24]. The algorithm still has quadratic storage and potentially substantial repeated silhouette cost; the changes reduce avoidable allocations rather than changing its asymptotic scaling.

## 3.7. STEP 2: Wasserstein consensus

Only acceptable local clusters enter consensus. For clusters $A$ and $B$, the implemented source metric is

$$D(A, B) = \left[ \sum_{f=1}^{4} W_1(A_f, B_f)^2 \right]^{1/2}. \tag{3.1}$$

Here $W_1$ is the one-dimensional first Wasserstein distance between empirical marginal descriptor distributions [25]. Equation 3.1 does not describe joint four-dimensional optimal transport. It can miss changes in dependence structure that leave every marginal distribution unchanged.

The reproduction configuration uses raw descriptors, matching the supplied code. This mixes descriptor scales and units in the aggregate norm; the paper does not fully resolve that scaling choice. The normalized alternative is retained as an explicit extension. Average-linkage agglomeration is followed by the source's cubic-spline elbow heuristic: smoothing parameter 5, with the maximum second derivative sought in the first 90% of merge steps. Fewer than five local clusters or an entirely flat merge-height curve yields an explicit failure instead of an invented cut.

## 3.8. STEP 3: final frame assignment

Each global centroid is the mean normalized feature vector of all accepted local-cluster members assigned to that family. All analysis-window frames, including residuals and frames outside reduced discovery groups, are assigned once to the nearest centroid. Distance ties use deterministic label ordering. Centroids are not iteratively refitted after reassignment: doing so would introduce a k-means refinement absent from the documented baseline.

The procedure partitions the chosen descriptor space. It does not establish kinetic metastability or atomic structural uniqueness. Final labels are stored with original frame identities so that downstream statistics, structures if later supplied, and stability comparisons can refer to precisely the same observations.

## 3.9. FES validation and interpolation

The FES parser requires explicit axis names, extrema, grid counts, and nonperiodic status. It verifies row count, finite values including supplied derivative columns, coordinate spacing, unique Cartesian pairs, and agreement with the header. Row order is not assumed. A partial-download extension is rejected before parsing.

Interpolation uses SciPy's linear rectilinear-grid interpolator with bounds checking enabled [26, 27]. Every query must fall inside both axes. A bilinear analytic surface containing a mixed $\alpha\beta$ term provides an exact test, including boundary queries and shuffled rows. Missing cells and out-of-bounds queries are errors; no extrapolation or clipping is permitted.

## 3.10. Statistical weights

Equation 2.2 is evaluated in log space. Within replica $r$, normalized weights are

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \mathrm{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \frac{\widetilde{w}_{ri}}{3}. \tag{3.2}$$

Equation 3.2 gives each replica equal total mass, independently of the arbitrary log-weight offset or differences in raw weight sums. The recorded raw exponential is diagnostic only. If it is not representable,

the output includes an explicit overflow/underflow status; normalized calculations remain based on log weights.

The intended frame table contains replica, source row, file line, frame index, time, physical CVs, interpolated free energy, raw log weight, raw weight/status, within-replica weight, pooled weight, and final label. No such real weighted table is reported in this release because the FES gate has not passed.

## 3.11. Weighted cluster characterization

Cluster relevance is the sum of pooled weights belonging to the cluster. Descriptor means and widths use the same weights restricted to cluster members. The width is the square root of the weighted central second moment, a measure of distribution spread rather than an unbiased estimator of sampling uncertainty. Cluster relevance is also calculated separately in each replica before reporting its mean and replica SEM.

The baseline bootstrap resamples contiguous observation blocks independently within each replica, renormalizes the resampled weights, and recomputes cluster populations. It is conditional on a fixed fitted partition and a fixed supplied FES. These intervals would not include uncertainty from FES reconstruction, model choice, or clustering across seeds. Those limitations are retained in the output metadata.

## 3.12. Software environment and reproducibility

The modular implementation uses NumPy, SciPy, pandas, Matplotlib and scikit-learn; the key numerical and machine-learning libraries have established primary descriptions [27–29]. A locked environment records installed versions separately from published simulation versions. Snakemake expresses stage dependencies, while CLI commands provide the same functionality without notebooks [4, 30].

Each run begins with a manifest and resolved parameters. A supervisor records the worker's status and samples process-tree memory every 0.1 s, enforcing configured runtime and memory limits. Successful, failed, and blocked runs have distinct final statuses. Figures may be rebuilt only from a successful manifest after its output hashes are checked. This implements a concrete provenance link rather than treating a familiar-looking plot as evidence that the intended calculation ran.

# 4. Reproduction and scientific code audit

## 4.1. Original implementation

The supplied implementation is a monolithic script whose top-level execution reads inputs, writes segmented replica files, creates figures, performs clustering, and computes summary tables. Static parsing identifies an un-commented Italian sentence at line 437, causing a syntax error before the analysis can run. The original is preserved exactly, including this defect; repairing it in place would destroy the evidence being audited.

Static inspection is sufficient to identify several intended operations, but it cannot establish that any previous full run succeeded. No original execution manifest or numerical output set was located. The present audit therefore distinguishes observed source behavior from observed runtime results. Where the source contradicts the published description, the modular implementation follows the explicitly documented scientific correction rather than reproducing an error for its own sake.

## 4.2. Identified methodological issues

The original parser silently takes the first three reset segments and skips certain malformed lines. The new parser checks segment count and row validity. Original group generation depends on a fixed list of 17 seeds, which would be exhausted under smaller group sizes. The replacement uses local RNG streams and retains a complete identity map. This is a reproducibility improvement but changes exact sampled subsets, so bitwise equivalence to the legacy script is not claimed.

In the clustering stage, rejected frames receive a generic label, and the consensus construction includes every label. This allows residuals that did not pass the acceptance criteria to affect the global families. The corrected implementation stores accepted memberships explicitly and keeps residuals out of centroid discovery. It subsequently assigns those residuals in the published STEP-3 operation.

The FES reader ignores completeness metadata, permits missing grid values with a warning, and the weight function uses nearest-grid lookup with boundary selection. These behaviors can generate plausible but unsupported weights. The replacement rejects incomplete grids and performs bounded bilinear interpolation. The original partial download ended inside a numerical row and was rejected. The complete replacement received on 7 October passed grid validation; it did not retroactively make the partial file usable.

Finally, the original weighted-analysis section calculates arithmetic descriptor means and standard deviations while weighting only relevance. This conflicts with the weighted characterization described in the paper's figure captions. Global pooled weights also do not guarantee equal replica contribution. Both issues are corrected explicitly. Distribution width, replica SEM and bootstrap uncertainty are separate outputs because they answer different questions.

**Scientific audit note**

Correcting a code path does not establish the validity of its input data or the convergence of the original simulation. The complete grid now passes validation, but its energy-unit and construction provenance still require clarification before empirical reweighting.

## 4.3. Scientific V2 corrections: evidence unavailable

The requested intermediate file `CLUSTERING40_scientific_v2.py` and its changelog were not located. The method map therefore marks its implementation column as unavailable throughout. None of the corrections in this report is attributed retrospectively to that version. If it is later supplied, it should be hash-verified, preserved without modification, and compared with both the original and the modular implementation.

## 4.4. Verification and numerical tests

The current suite passed 35 tests, including strict parsing, interval boundaries, normalization, analytic bilinear interpolation, incomplete and duplicate grids, finite/bounds checks, weight sign, free-energy-offset invariance, equal replica masses, weighted moments, residual handling, final assignment, label-permutation agreement, and bootstrap separation of replicas. A seeded autoregressive fixture checks that the correlation estimator behaves plausibly for a known stationary process.

A complete synthetic pipeline fixture exercises cluster discovery, consensus, all-frame assignment, final-bias weights, population tables, conditional bootstrap intervals and PDF generation. Its generated descriptors and analytic surface are confined to testing and are not treated as Aβ40 observations. An additional repeatability test fits the GMM extension to explicitly synthetic separated observations. Its label-permutation and missing-family tests verify comparison behavior. These tests establish numerical behavior, not the validity of a biological family.

## 4.5. Remaining methodological ambiguities

The main unresolved choices are the exact STEP-2 descriptor scaling, the construction of the replica-averaged FES, and the detailed convention for published cluster widths and uncertainties. Density ties and spline-based elbow selection are also more explicit in the supplied code than in the paper. The repository records these as source-informed choices rather than presenting them as fully specified published theory.

The distinction is operational: raw Wasserstein scaling remains the reproduction setting, while normalized scaling receives a different extension configuration. A missing FES-generation recipe remains a provenance limitation even after a complete numerical grid is obtained. A future comparison with published populations must assess those differences before attributing a discrepancy to molecular behavior.

# 5. RobustCluster-Aβ: robustness extension

## 5.1. Temporal correlation and effective sampling

For each replica and baseline descriptor, the extension computes a demeaned autocorrelation sequence using an FFT-based convolution. Covariances use the common $1/N$ normalization. Adjacent positive-lag terms are paired; the sum is truncated at the first nonpositive pair and monotonized by cumulative minima. The resulting diagnostic is

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5.1}$$

In eq. (5.1), $\mathscr{K}$ denotes the retained lag contributions after the positive-sequence processing; the implementation saves the truncation lag. The convention includes the zero-lag half term in $\tau_{\mathrm{int}}$. Different conventions in the literature differ by that term, so reporting only a number called "correlation time" would be ambiguous.

The method flags an estimate when the sequence has no observed truncation, when $N < 50g$, or when the absolute difference between first- and second-half means exceeds 0.25 of the full-series standard deviation. These are explicit screening heuristics, not a formal stationarity test. Their role is to discourage false precision in a biased, possibly drifting time series. Sampling-quality literature emphasizes that uncertainty estimates depend on the adequacy of the data and assumptions, rather than on the existence of a formula alone [11, 12].

Weight concentration is assessed separately by

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{5.2}$$

Equation 5.2 describes how unequal weights reduce the effective number of contributing observations, assuming no additional dependence correction. It is not multiplied by, or automatically divided by, the correlation ESS. A single combined ESS would require an estimator-specific treatment of dependence in the weighted observable. No real Kish ESS is reported before validated statistical weights exist.

## 5.2. Cluster stability across seeds

The planned DPC seed experiment reruns random grouping, iterative DPC, consensus cutting and final assignment. Comparing only a fixed clustering with different label integers would not test discovery stability. The smoke profile uses two seeds, a separate pilot uses five, and the held full profile defaults to 20, with configuration support for a larger 20–50-seed study.

Partitions are compared on identical source-frame identities. ARI assesses agreement independent of label names [31]. A maximum-overlap assignment then matches families across runs, after which Jaccard overlap, population variation and centroid movement can be summarized. Unmatched families are retained explicitly, including the appearance or disappearance of a family. A small average variation among only the matched families would otherwise conceal instability in cluster number.

Centroid comparisons across representations use the original four descriptors, rescaled by the common baseline ranges. Thus a PCA-space centroid is not compared directly with a raw-space centroid. Population variation across seeds describes algorithmic sensitivity under the same input ensemble; it is not interchangeable with sampling uncertainty between independent replicas.

## 5.3. Feature redundancy and PCA

The completed lightweight diagnostics calculate the pooled Pearson correlation matrix and PCA after standardizing each baseline feature to zero mean and unit variance. The explained-variance spectrum

measures the geometry of the observed descriptor cloud. It does not measure thermodynamic relevance or identify a slow physical mode.

The alternative clustering configurations retain all components in standardized PCA space, with or without whitening. Retaining all components makes PCA an orthogonal rotation of standardized variables; the rotation alone leaves Euclidean distances unchanged. Any difference from the min–max baseline therefore concerns standardization, whitening, numerical effects, or downstream choices, not information loss from dimensionality reduction. This prevents an apparent PCA benefit from being misdescribed as discovery of a reduced kinetic model.

## 5.4. Independent GMM feature-ablation experiment

The completed original experiment compares the four baseline descriptors with $n_\alpha$, $n_\beta$ and $R_g$ after removing SASA. It uses Gaussian mixtures in scalar descriptor space, a different model from both the published DPC pipeline and coordinate-based ShapeGMM. For a standardized observation $\mathbf{x}$,

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{5.3}$$

Equation 5.3 approximates the sampled descriptor distribution. Its mixture coefficients are not equilibrium statistical weights. The implementation uses scikit-learn [29]; full covariance matrices allow dependencies between features within a component.

A fixed random fit sample contains 1,000 frames from each replica. Sampling uses seed 20261007; feature means and standard deviations are estimated on those 3,000 rows only. The same observations and shared-feature scaling are retained for both representations. Five initialization seeds (42–46) are used at each $K = 2, \ldots, 6$, with one initialization per fit, a 500-iteration limit, tolerance $10^{-3}$ and covariance regularization $10^{-6}$. One initialization deliberately exposes local-solution sensitivity. The protocol was committed before execution.

Every model assigns all 300,003 observations, including the fit rows. This is a descriptive sensitivity analysis, not a held-out test. ARI compares the two representations at each seed and component count; all ten seed pairs are also compared within each representation. Maximum-overlap matching records corresponding labels, Jaccard overlaps, unmatched families and centroid shifts in the shared standardized three-feature space. Raw label fractions are retained separately by replica. Observed ranges across runs are not confidence intervals.

This experiment holds the fit subset fixed to isolate representation and initialization effects. It does not quantify sample-selection uncertainty or remove temporal dependence. BIC is stored for comparisons within a feature space only; it is not compared across dimensions or used to declare a physical number of states. The planned DPC ablations remain separate experiments with their own provenance requirements.

## 5.5. Wasserstein scaling sensitivity

The raw consensus distance and the normalized alternative use the same marginal-distance construction, changing only the descriptor scaling used in that stage. This isolates a specific implementation ambiguity. Because raw SASA and radius-of-gyration values have different units and numerical ranges, an unscaled aggregate norm can be influenced disproportionately by the larger-range variable.

The normalized result would not establish which version the original authors intended. It would show how sensitive the downstream families and populations are to the chosen convention. Joint distances, sliced Wasserstein, energy distance and MMD are possible later studies, but none is implemented as an undisclosed replacement for the current baseline.

## 5.6. Reweighting sensitivity

The optional comparison holds the fitted partition fixed and contrasts the final-bias estimator with weights derived from verified `metad.rbias`. It compares reweighted distributions of all four baseline

descriptors and family relevance. This separation attributes any differences to weighting rather than to simultaneous changes in the partition.

The descriptor input contains an appropriately named column, but the original PLUMED configuration and energy-unit evidence are absent. Therefore the optional comparison is blocked. Its theoretical motivation and software interface exist; an empirical agreement or disagreement is not claimed. Reweighting alternatives are discussed in the primary estimator literature and official PLUMED reference [16–18].

## 5.7. Alternative density-based clustering

DPA is a high-value future benchmark because it provides a different treatment of density estimation and significant peaks [21]. Its evaluation should compare family number, location, population and correspondence using the same valid observation identities and weights. Installing a dependency would not by itself constitute that benchmark; a documented fit and output set are required.

Weighted ShapeGMM is relevant to biased structural ensembles [22]. However, applying it as a coordinate-based comparison requires atomic structures, a consistent atom selection and alignment convention, and valid frame mapping. Scalar descriptor rows cannot be expanded into those structures. The repository therefore leaves this option explicitly deferred.

## 5.8. Optional external validation

The experiment-constrained Aβ40 ensemble literature offers an opportunity to compare global compaction and secondary-structure behavior with an independent source [7]. Before numerical comparison, sequence, temperature, weighting, atom selections and observable definitions must be checked. A mass-weighted $R_g$ and a Cα-only $R_g$, for example, need not have directly comparable distributions.

This release does not download and process an external ensemble or claim agreement with NMR/SAXS measurements. The feasibility investigation is limited to identifying the relevant study and compatibility requirements. External validation remains a distinct stretch goal rather than a narrative substitute for missing local reproduction results.

# 6. Results

## 6.1. Available evidence and execution status

The real descriptor table passes the implemented schema, finite-value, replica and interval checks. Each of the three segments contains 100 001 rows at the source-labelled 20 ps spacing. Thus the data are suitable for the reported scalar-descriptor diagnostics, subject to the provenance assumptions described earlier. This validation does not independently establish simulation convergence or the correctness of upstream molecular definitions.

The reported diagnostics come from run `20260921T011959-583b4421`. Its recorded elapsed time is 2.28 s, and its sampled peak worker-process memory is 418.2 MiB. These measurements describe this lightweight local run only. They are not estimates of full DPC runtime, nor evidence of HPC execution. The earlier reproduction attempt stopped at the missing-FES gate. Subsequent complete-grid validation and the independent GMM run have distinct manifests and do not convert that blocked attempt into a successful reproduction.

| Analysis | Evidence status |
|---|---|
| Input validation | Completed on the supplied Aβ40 descriptor table |
| Correlation matrix / PCA | Completed; unweighted biased-sample diagnostics |
| Autocorrelations / provisional ESS | Completed; all reliability flags failed |
| FES grid and interpolation | Completed; units/construction remain unverified |
| Weighted paper reproduction | Blocked: unit/construction provenance |
| GMM seed / SASA-ablation pilot | Completed; biased-sample geometric comparison |
| DPC seed / feature / scaling studies | Not completed; separate reproduction prerequisites |
| Time-dependent weight comparison | Blocked: FES and rbias unit/provenance requirements |
| Molecular representatives / ShapeGMM | Blocked: coordinate trajectory and mapping absent |
| External validation / Leonardo jobs | Not performed |

**Table 1.:** Scientific evidence available at release. Tests on synthetic fixtures are excluded from the empirical analysis statuses.

## 6.2. Descriptor distributions and redundancy

The pooled Rg–SASA Pearson correlation is 0.848, whereas the correlation between the two secondary-structure descriptors is -0.456. These are properties of the biased observation sample. The strong compaction/exposure correlation motivates a SASA-ablation experiment, but does not demonstrate that either variable is dispensable for every local family.

The first standardized principal component explains 51.8% of total variance, and the second explains 36.3%, giving 88.1% cumulatively. Figure 3 presents the numerical correlation matrix and spectrum. This concentration of variance supports examining redundant directions; it does not justify discarding a lower-variance variable without checking the effect on clustering and relevance.

Figure 4 shows per-replica empirical densities on common descriptor-specific bin edges. Each histogram is normalized by its own frame count and bin width. Differences between replicas are visible, but their equilibrium significance cannot be determined from these raw distributions alone. The figure is intentionally labelled as biased-sample density to avoid conflating observation frequency with unbiased relevance.

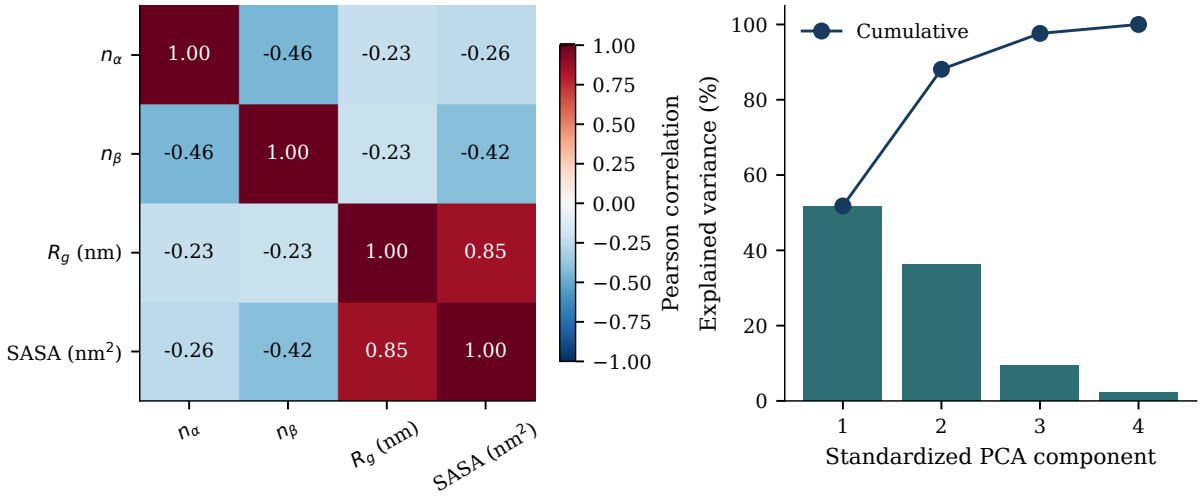



**Figure 3.:** Computed diagnostics for the pooled, unweighted Aβ40 descriptor sample. Left: Pearson correlations of the four baseline descriptors. Right: explained variance of PCA after per-feature standardization, with cumulative variance shown by the line. These are geometric summaries of the biased sample and do not represent reweighted equilibrium probabilities.

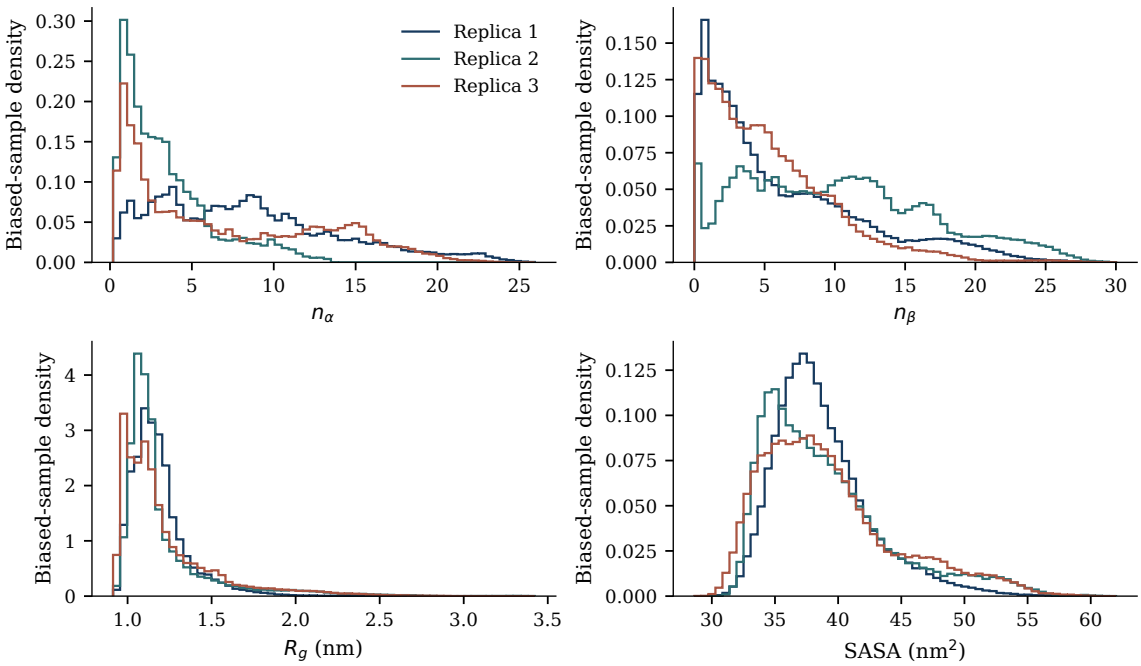



**Figure 4.:** Actual per-replica descriptor histograms over the supplied interval. Curves use common bin edges for each descriptor and independent density normalization. No final-bias or time-dependent statistical weights have been applied. Units follow the original script's labels and remain a recorded upstream-provenance assumption.

## 6.3. Temporal correlation and provisional ESS

The correlation diagnostics produce provisional ESS values ranging from 3.2 to 28.7 across the 12 replica–descriptor series, compared with 100 001 raw frames in each series. All 12 estimates fail at least one configured reliability heuristic. They should therefore be read as warning indicators of strong dependence and possible nonstationarity, not as precise counts of independent equilibrium conformations.

Table 2 reports the estimator outputs together with the half-window mean shift. Several descriptors exhibit substantial differences between the first and second halves of the recorded interval. The extension does not convert those shifts into a formal rejection of the published convergence analysis, because the full bias history and corresponding convergence diagnostics are unavailable. It establishes that naive independent-frame uncertainty estimates would be poorly supported by the supplied stream.

| Replica | Descriptor | $g$ | $\tau_{\mathrm{int}}$ (ns) | $N_{\mathrm{eff,corr}}$ | Half shift / SD |
|---|---|---|---|---|---|
| 1 | $n_\alpha$ | 22067 | 220.7 | 4.5 | 1.28 |
| 1 | $n_\beta$ | 11079 | 110.8 | 9.0 | 0.83 |
| 1 | $R_g$ | 3679 | 36.8 | 27.2 | 0.27 |
| 1 | SASA | 3490 | 34.9 | 28.7 | 0.36 |
| 2 | $n_\alpha$ | 6259 | 62.6 | 16.0 | 0.37 |
| 2 | $n_\beta$ | 10843 | 108.4 | 9.2 | 0.41 |
| 2 | $R_g$ | 3951 | 39.5 | 25.3 | 0.33 |
| 2 | SASA | 5325 | 53.2 | 18.8 | 0.17 |
| 3 | $n_\alpha$ | 31742 | 317.4 | 3.2 | 1.61 |
| 3 | $n_\beta$ | 5880 | 58.8 | 17.0 | 0.71 |
| 3 | $R_g$ | 6259 | 62.6 | 16.0 | 0.88 |
| 3 | SASA | 9958 | 99.6 | 10.0 | 0.87 |

**Table 2.:** Computed correlation diagnostics. All entries are flagged as unreliable under the stated heuristics. The time scale is measured along the biased trajectory, using the source-labelled sampling interval. $g$ is statistical inefficiency; the final column is the absolute difference of half-window means divided by the full-series standard deviation. No physical relaxation-time interpretation is made.

## 6.4. Complete FES validation

The recovered file contains exactly 2,801 by 1,301 grid points: 3,644,101 rows with five finite columns, no duplicate coordinates and complete Cartesian coverage. Its nonperiodic axes span 0–28 and 0–13 at spacing 0.01. All 300,003 converted descriptor queries lie within those bounds, and bounded bilinear interpolation returns finite values for every query. The grid check is an input-integrity result; the unlabeled energy column is not assigned a unit solely because the source script assumes kJ/mol.

## 6.5. Representation and initialization sensitivity

The GMM pilot fitted 50 models, all reporting convergence, in 7.01 s with a sampled process-tree peak of 660.2 MiB. Convergence here means that the configured numerical stopping criterion was met, not that every model found the same or globally optimal solution. The comparison in fig. 6 reveals substantial differences between component counts. At three components, cross-representation ARI lies between 0.862 and 0.868, with approximately 95.3% of frames retaining corresponding labels after maximum-overlap matching (median across seeds). Within-space seed agreement is also high: 0.994–1.000 with four descriptors and 0.988–1.000 without SASA. At two components, however, within-space ARI ranges down to 0.011 and 0.067, respectively. A single initialization would conceal this variability. The full matrices are retained in fig. 7.

The result supports a qualified statement: feature redundancy is conditional on representation and model resolution. SASA removal preserves much of the three-component partition in this pilot, but has different consequences at other component counts. No weighted relevance, equilibrium population or

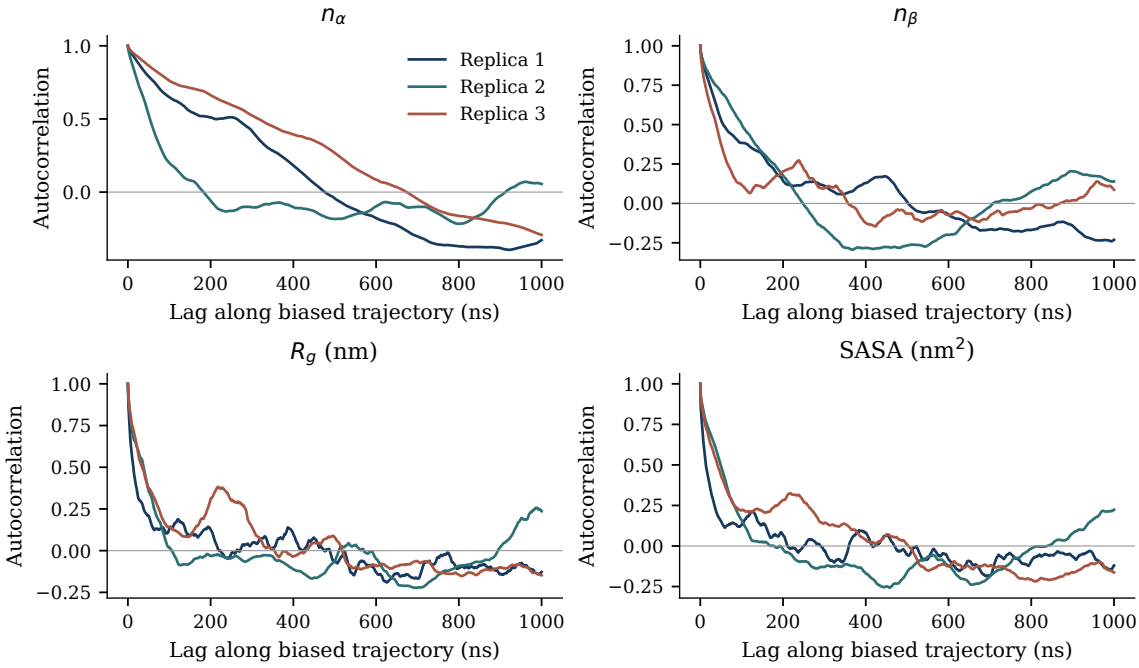



**Figure 5.:** Autocorrelation functions computed separately for each replica and descriptor. The plotted curves cover the first half of the available lag window with a reduced plotting density; the estimator operates on the full, un-decimated ACF sequence. Long persistence and changing behavior across replicas reinforce the need for cautious uncertainty analysis. Lag is biased trajectory time, not recovered physical time.

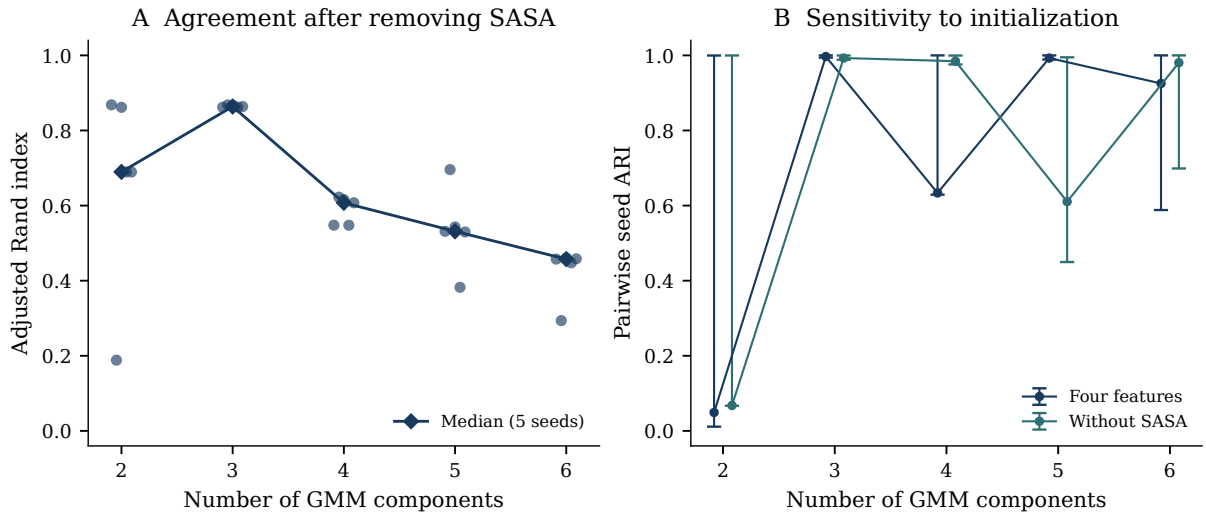



**Figure 6.:** Independent GMM sensitivity experiment on the biased descriptor sample. Left: four-feature versus SASA-ablated assignments at each of five seeds; diamonds mark medians. Right: within-space ARI over ten seed pairs; bars span observed minima and maxima, not confidence intervals. The fit subset is fixed across all models and evaluation includes fit rows.

| Components | Median ablation ARI | Observed range |
|---|---|---|
| 2 | 0.690 | 0.188–0.868 |
| 3 | 0.864 | 0.862–0.868 |
| 4 | 0.607 | 0.548–0.622 |
| 5 | 0.531 | 0.382–0.696 |
| 6 | 0.457 | 0.294–0.461 |

**Table 3.:** Observed representation agreement across five paired initializations. Component count was varied by design; the table does not identify an optimal biological family count.

physical transition rate follows from these comparisons.

## 6.6. Analyses remaining unavailable

Equilibrium cluster relevance, weighted family characterization and completed paper-DPC robustness remain unavailable. No molecular snapshots are created because the descriptor table has no atomic coordinates. The completed GMM experiment is an original descriptor-space extension, not a replacement for those stages.

# 7. Discussion

## 7.1. What the completed experiments support

The results show why a representation choice should be tested rather than decided from a correlation matrix. Rg and SASA are strongly associated globally, yet agreement after SASA removal varies with GMM component count. At three components, the two partitions retain high agreement and are stable across the tested initializations. At other counts, representation or initialization has a larger effect. These findings concern one fixed fit subset and one model class; they neither establish that SASA is indispensable nor justify removing it from the published baseline.

The temporal diagnostics qualify a different assumption. Hundreds of thousands of saved frames need not provide equally many independent observations. Here all correlation-estimator reliability checks fail, so the provisional ESS values are warning indicators rather than calibrated measures of ensemble size. Random selection of fit frames does not remove this dependence. New fit subsets and time-block comparisons would be needed to assess sampling sensitivity [12].

## 7.2. Relation to the published study

The repository reconstructs the published analysis design, but has not reproduced its reweighted conformational families or their populations [2]. The completed GMM pilot is an independent descriptor-space extension. It provides a methodological comparison, not validation of published biological conclusions. Neither the diagnostic correlations nor stable Gaussian assignments establish aggregation rates, toxicity or metastable states. Comparison with an independently constrained ensemble would additionally require compatible structures and observable definitions [7].

## 7.3. Limitations and methodological implications

Numerical convergence does not establish a global optimum, and partition stability does not identify a unique physical state model. Mixtures approximate a biased descriptor distribution; marginal Wasserstein distances in the separate baseline can miss joint dependence. Representative structures, contact patterns and alternative feature definitions would help evaluate these limitations, but cannot be recovered from scalar rows alone.

The recovered FES resolves an input-completeness problem. It does not establish the producing units, replica combination or convergence of the bias. Reweighting would introduce weight concentration as another source of variability, to be assessed separately from temporal dependence and initialization sensitivity. With only three replicas, uncertainty in replica-level comparisons would also need careful interpretation.

Reproducibility makes these distinctions inspectable. Configurations expose assumptions, tests expose numerical failures, and manifests distinguish successful from blocked runs. The practical improvement is not simply deterministic execution: it is the ability to trace an assertion to its inputs and estimator. Transparent code supports scientific evaluation, while leaving the validity of the simulation and physical interpretation open to evidence [6, 32].

# 8. Internship outcomes and skills

## 8.1. Learning sequence and practical objectives

The internship connected the Data Analysis degree with molecular research. The documented learning programme began with amino-acid properties, peptide bonds, protein structure and intrinsic disorder. It then introduced force fields, molecular dynamics and dihedral angles through supplied slides and reading materials. The final-stage assignment asked for reproduction of the supplied clustering analysis, investigation of parameter changes, and an independent analysis using methods from the degree. These describe the programme and its objectives; they are not a reconstructed attendance or hours log.

The project applies statistical ideas to a setting where the observations require physical interpretation. A row is a saved configuration, but consecutive rows can be strongly correlated. A geometric cluster is a useful summary, but its frequency in a biased simulation is not automatically its equilibrium probability. Keeping those distinctions explicit was central to connecting data analysis with the molecular problem.

## 8.2. Demonstrable analysis and software outputs

The completed outputs include an input and code audit, modular numerical routines, 35 passing tests, real descriptor diagnostics, a complete-grid FES check, and an independent GMM sensitivity experiment. The latter turns a descriptive observation—the Rg–SASA correlation—into a testable question about assignments. Recording all component counts and initializations prevents the discussion from depending only on a favorable plot.

The implementation also makes a common software decision scientifically visible: scaling is fitted on specified observations, then applied consistently. The following excerpt is taken directly from the analysis source and remains searchable in the PDF.

**01 Fit the geometry once** RobustCluster-Aβ / Python

```python
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

This function rejects a constant fit feature. A numerical test checks that a large evaluation value does not alter the fitted mean and scale. Its purpose is to make the geometry used by the model reproducible, rather than leave normalization as an undocumented preprocessing step.

## 8.3. Companion projects: VMD MCP and GROMACS MCP

The student identifies two public software projects as additional outputs: VMD MCP and GROMACS MCP. Their inspected source provides concrete examples of interfaces around molecular software. The Model Context Protocol (MCP) exposes operations with structured inputs and outputs to a client; the scientific calculation still depends on the underlying program and supplied data.

VMD MCP constructs Tcl commands for operations such as per-frame radius of gyration, RMSD, SASA and rendering. The gyration function passes the selected atoms and mass-weighting option to VMD, then parses frame-indexed values. GROMACS MCP includes helpers for simulation preparation and analysis and a background-run interface that constructs a command, validates paths and records output. Genuine source screenshots and immutable commit links appear in figs. 8 and 9.

These repositories complement RobustCluster's descriptor-analysis layer, but they are not asserted to have generated the supplied Aβ40 ensemble. In particular, the inspected VMD gyration function returns

values in its documented angstrom convention; comparisons with source-labelled nanometre descriptors require explicit unit checks. The repository histories retain contributor attribution. Their interfaces are software outcomes; scientific validation still depends on recorded calculations and compatible inputs.

## 8.4. Reproducibility and computational scope

The practical software skills represented here include preserving source inputs, separating configuration from implementation, testing analytic examples, handling numerical failures, tracking versions and producing figures from saved tables. Manifests connect these practices to scientific interpretation: they identify which calculation supports a claim and which assumptions remain unresolved.

All newly reported calculations are local descriptor analyses. The project includes a parameterized Leonardo submission template and official documentation references [33], but no Leonardo job records exist. HPC execution, molecular simulation setup and physical kinetic measurements are therefore outside the claimed computational outcomes.

# 9. Conclusions and future work

RobustCluster-Aβ converts the available internship materials into an auditable computational project while retaining a clear limit on its scientific claims. The supplied descriptor data support validation, feature diagnostics and temporal-correlation screening. The source audit identifies concrete issues in execution, residual-cluster handling, FES interpolation, replica pooling and descriptor weighting. The modular implementation addresses those issues and supplies numerical and integration tests without modifying the original files.

The real diagnostics show strong Rg–SASA correlation, a concentrated standardized PCA spectrum, and unreliable correlation-time estimates under the configured heuristics. The completed GMM ablation demonstrates that representation agreement depends on component count and initialization; the correlation result alone cannot establish feature redundancy. The temporal diagnostics caution against independent-frame uncertainty formulas. They do not establish equilibrium populations, kinetic rates, or a completed reproduction of the published conformational families.

The complete FES now passes grid and query validation. The next reproduction step is to establish its energy units and replica-combination provenance. With those requirements satisfied, the baseline must pass smoke execution before resource estimates are used to authorize larger runs. A production result should retain the published scientific parameters and explicitly distinguish any source-informed implementation choices. Seed, feature and Wasserstein experiments can then quantify the consequences of those choices on common frame identities.

Verified bias-offset information would enable the optional time-dependent reweighting comparison. Atomic trajectories and a valid indexing map would enable representatives and coordinate-based benchmarks using tools such as MDAnalysis and ShapeGMM [22, 34]. External experimental-ensemble validation should proceed only after observable definitions and data rights have been checked.

The present result is therefore a reproducible scientific foundation and an incomplete empirical reproduction. Its usefulness rests on making that distinction testable: successful outputs have provenance, unavailable analyses have explicit blockers, and future claims require actual computations rather than changes in wording.

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# A. Full parameter table

| Parameter | Value | Evidence / role |
|---|---|---|
| Replicas | 3 | Primary paper; observed table segmentation |
| Analysis interval | 1–3 µs | Published baseline; inclusive source-labelled ps boundaries |
| Temperature | 310 K | Published weighting parameter; no new MD run |
| Bias factor $\gamma$ | 10 | Published weighting parameter |
| Molar gas constant | 0.008314462618 kJ $mol^{-1}$ $K^{-1}$ | Implementation unit convention |
| CV conversions | $\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$ | Published descriptor scaling |
| Baseline features | $n_\alpha$, $n_\beta$, $R_g$, SASA | Pooled min–max normalization |
| Initial $d_c$ | 1.3 | Published baseline |
| Cutoff multiplier | 0.9 | Published 10% reduction |
| Center-count scan | 10 down to 2 | Published baseline |
| Minimum cluster size | 300 frames | Published acceptance criterion |
| Minimum mean silhouette | 0.3 | Published acceptance criterion |
| Failure stopping count | 5 | Consecutive iterations with no accepted cluster |
| Distance storage | float32 | Source-informed; blockwise construction |
| Consensus | Average linkage | Marginal Wasserstein norm |
| Elbow smoothing | $s = 5$ | Supplied source heuristic |
| Elbow search range | First 90% of merges | Supplied source heuristic |
| Final assignment | One nearest-centroid pass | No iterative refitting |
| Master seed | 42 | Default project configuration; recorded per run |
| Smoke / medium groups | 3,000 / 10,000 | Reduced execution profiles |
| Smoke / medium discovery groups | 2 / 3 | Not full-production evidence |
| Full group size | About 30,000 | All groups; full execution disabled |
| Seed counts | 2 smoke; 5 pilot/medium; 20 full default | Full robustness configurable; not performed |
| Bootstrap draws | 500 | Conditional moving-block extension; not performed on real weights |
| Resource policy | Bounded sequential local workers | Full execution held for review |

# B. Mathematical derivations and estimator checks

## Why the sign of the final-bias correction is negative in FES form

Under a static bias $V$, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in eq. (2.2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

## Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica $r$. The pooled expectation of $A$ is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses $A$ equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

## Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0, 1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and $uv$. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

# C. Additional diagnostics and interpretation

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a plotting subsample, not the series used to estimate $g$. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

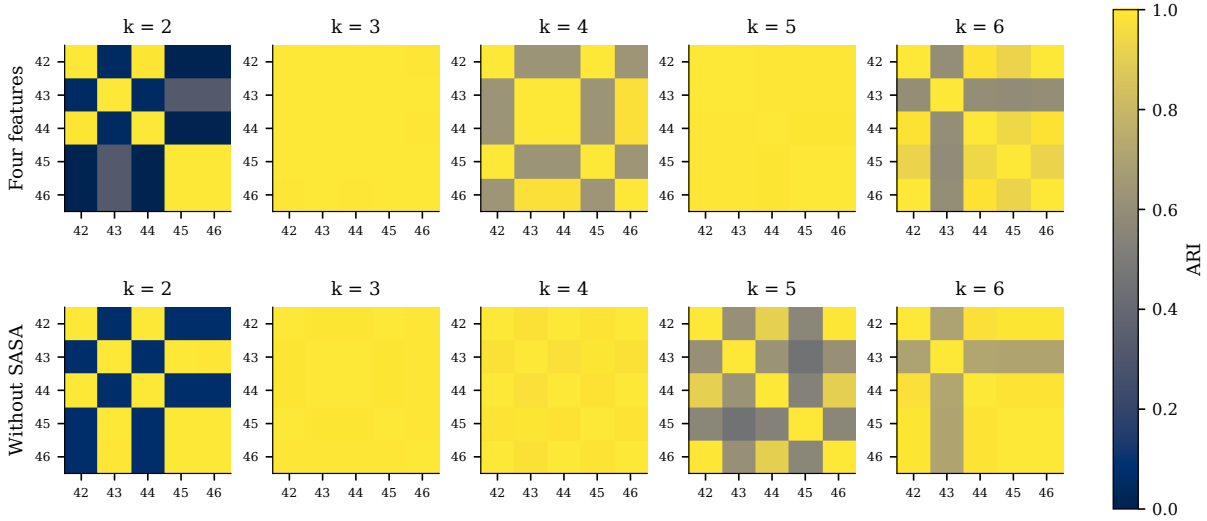



**Figure 7.:** All pairwise seed ARIs from the GMM pilot. Columns show each tested component count; rows distinguish the four-feature and SASA-ablated representations. The diagonal is identity. All fits converged, so no entries are omitted. Similar numerical stopping status can coexist with different partitions.

# D. Code audit summary

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

Codex assisted with software refactoring, numerical checks, document preparation and layout. Reported numerical observations come from recorded Python runs; the illustrated cover and two conceptual editorial artworks were generated with the built-in image-generation tool, and the source-code screenshots are actual browser captures. This assistance is separate from the published methodology and the original supplied materials. The generation tool does not expose a verifiable model-version identifier; no specific Image 2.5 execution claim is made. Exact prompts and image hashes are retained in `docs/ARTWORK_PROMPTS.json` and `report/artwork_provenance.json`. The supplied inspiration image guided the print texture and restrained palette, not the scientific content. The cover typography is part of the generated image; its institutional and personal details were visually checked. The University of Messina logo is a separate image supplied by the student and placed without generative reconstruction.

# E. Reproducibility instructions

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The report imports successful diagnostics, FES-audit, GMM and figure manifests with matching hashes. Its generated editorial artwork and real browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

For later production, establish the recovered FES energy units and construction provenance, run smoke validation, review resource measurements, and create a reviewed full-run configuration. HPC templates require an actual allocation and explicit settings. Do not copy account or partition values from unrelated examples, and preserve job evidence before attributing calculations to Leonardo [33].

# F. Software and environment table

| Component | Version in diagnostic manifest |
|---|---|
| Python | 3.12.6 |
| numpy | 2.5.3 |
| scipy | 1.18.1 |
| scikit-learn | 1.9.1 |
| pandas | 2.3.3 |
| matplotlib | 3.11.2 |
| robustcluster-abeta | 0.1.0 |

**Table 5.:** Actual software versions for the reported local descriptor diagnostics. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.

# G. Companion source-code evidence

The following images are genuine browser captures from public, commit-specific GitHub source pages on 7 October 2026. They document implementation, not simulation execution. The full images are preserved unchanged; short searchable listings in the main text are a separate presentation format.

**02 Measure each saved frame** — VMD MCP / source lines 250–266

```
250      weight = "weight mass" if mass_weighted else ""
251      tcl = f"""
252  {_load_block(structure, trajectory)}
253  set sel [atomselect $mol "{_tcl_quote(selection)}"]
254  set n [molinfo $mol get numframes]
255  for {{set i 0}} {{$i < $n}} {{incr i}} {{
256    $sel frame $i
257    puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
258  }}
259  quit
260  """
261      res = _run_tcl(tcl)
262      series = [
263          {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
264          for m in res.get("markers", [])
265          if "rgyr" in m
266      ]
```

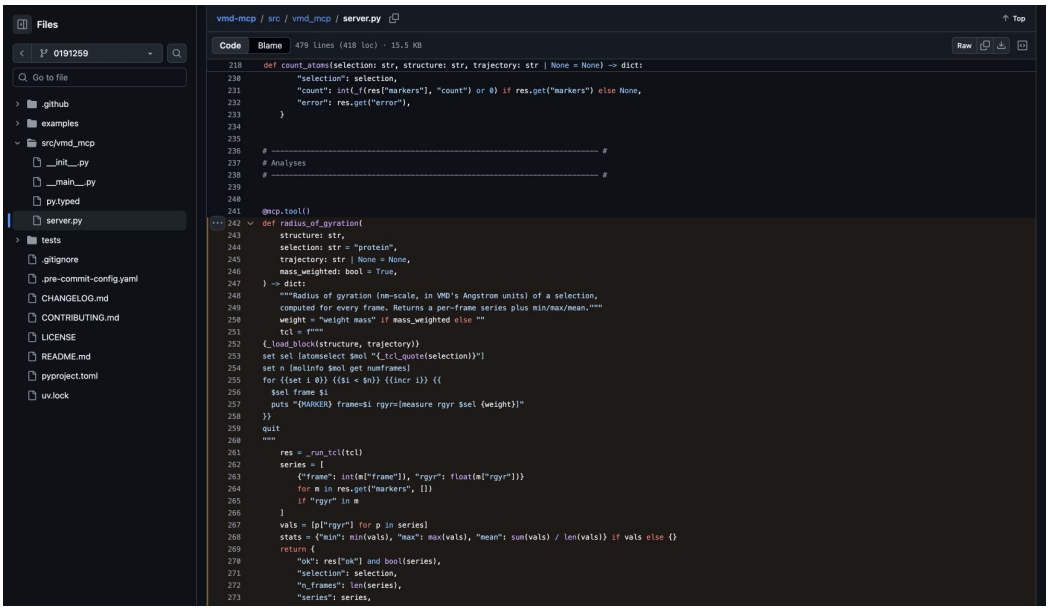



**Figure 8.:** VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. The function's documented unit convention must be checked before comparison with other tools. Open the inspected source.

**03 Validate before starting a job** GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

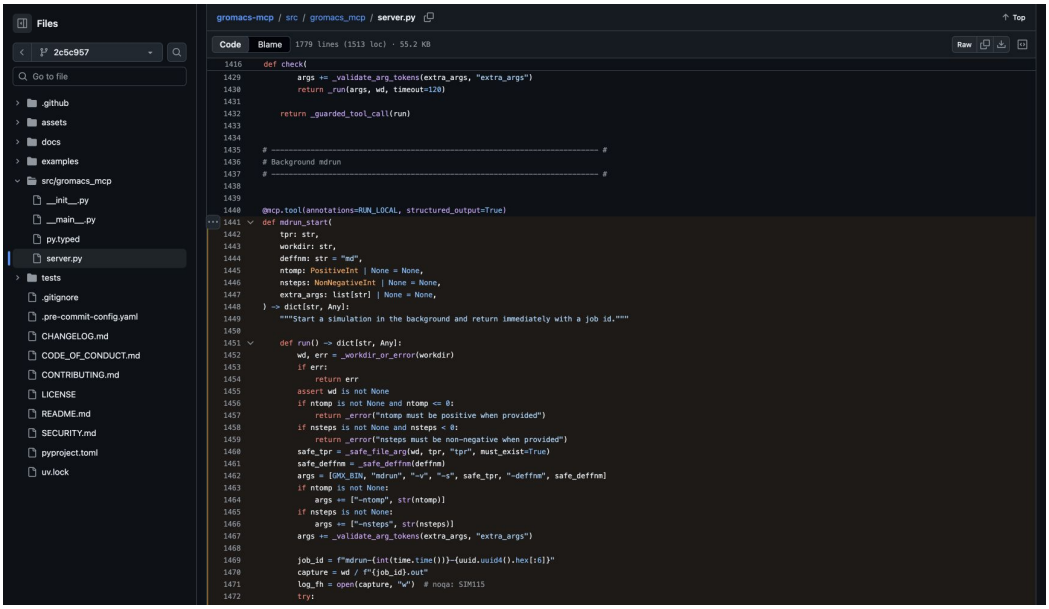



**Figure 9.:** GROMACS MCP source at commit 2c5c957: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. Open the inspected source.